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Effects of strength training with free weights and elastic resistance in older adults: A randomised clinical study

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ABSTRACT

Background: The aging process leads to negative changes in various bodily systems, including the neuromuscular system. Strength training, is considered the best strategy to counteract these neuromuscular changes, preventing sarcopenia and frailty in older adults.
Objective: To compare the effects of strength training with elastic resistance and free weights on the muscle strength of knee extensors and flexors and functional performance in the older adults.
Methods: This was a randomised clinical study. Thirty-one participants of both sexes were allocated randomly into two groups: Training Group Free Weight (TGFW, n = 15) and Training Group with Elastic Resistance (TGER, n = 16). Two individuals were excluded and so, twenty-nine individuals were evaluated before and after eight weeks training protocol, which was performed three times a week. The determination of the training load was obtained using a protocol of 10 repetitions maximum.
Results: No significant differences were found in either the intra- or the inter-group comparisons, on functional performance and peak muscle strength. In the intra-groups (pre- and post-strength training), it was observed that both groups significantly increased the training load (10 RM) for the extensors (TGFW $p = 0.0002$; TGER $p = 0.0001$) and the knee flexors (TGFW $p = 0.006$; TGER $p = 0.0001$).
Conclusion: Both training protocols similarly were effective in increasing the training load observed by the 10 RM test of the extension and flexion movements of the knee.

1. Introduction

Recent advances in geroscience have demonstrated that what were previously regarded as 'natural causes' or 'normal age factors' can now frequently be explained by an accumulation of potentially preventable deficits in various bodily systems, including the neuromuscular system (Sloane et al., 2019). The widespread and slowly progressive decrease in muscle mass and strength is characterized as the main factor for diagnosing sarcopenia, which is now considered a neuromuscular disease (Cao and Morley, 2016; Cruz-Jentoft et al., 2019) and is associated with negative outcomes, such as physical disability, lower quality of life, frailty, and higher mortality rates (Martinez et al., 2016).

Estimates of physical performance are important parameters for evaluating the health of the elderly, with gait speed being linked to lower limb strength and physical disability (Soares et al., 2019), which is highly relevant in clinical practice for screening and diagnosing severe sarcopenia (Cruz-Jentoft et al., 2019). Tests for measuring the capacity of physical performance represent a simple means of assessing gait, balance, agility, functional resources, and level of physical activity in the older adults (Viteckova et al., 2020). However, because of the growing number in the older adults population, the implementation of preventive and therapeutic interventions has become an additional challenge (Cruz-Jentoft et al., 2019).

Strength training (ST) has been demonstrated to have positive effects on health, and quality of life, and mainly as an intervention to improve the neuromuscular function in humans (Schoenfeld et al., 2019). In

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addition, it can currently be considered the best non-pharmacological intervention for aging and treating chronic diseases, as it directly influences the metabolic health of the skeletal muscle system (Abou Sawan et al., 2022; Fragala et al., 2019; Izquierdo et al., 2021; Severinsen and Pedersen, 2020). Strength training performed with free weights or on machines have constant resistance, and both are constantly applied in clinical practice, mainly because they present similar beneficial results (Mende et al., 2022; Teles et al., 2016). Despite being widely used due to their effectiveness and safety during practice, machines and free weights have a higher cost and can still cause apprehension for some individuals (Colado et al., 2020b; Martins et al., 2014).

On the other hand, elastic resistance tubes are also used as an alternative strategy for strength training and can be applied for all ages and with different tensions (light-medium-high), according to the level of physical conditioning of each participant (Özsu, 2018). Resistance band training is considered popular, being low-cost, and versatile, which ensures ease of movement execution in any direction, and efficiently contributes to rehabilitation and specific physical preparation for athletes (Seguin et al., 2022). The use of elastic training is not recent and has proven feasible for various populations (Seguin et al., 2022) such as athletes and young adults (Anderson et al., 2008; Shoepe et al., 2011), individuals with obesity (Liu et al., 2022), osteosarcopenic obesity (Banitalebi et al., 2020), and older adults (Caldo-Silva et al., 2023; Colado et al., 2020a; Martins et al., 2015).

Despite there being evidence demonstrating similarity in strength gains (Lopes et al., 2019a)and changes in body composition (Lopes et al., 2020) between elastic resistance training and traditional strength training, there are few studies that have compared the effectiveness of elastic resistance training with free weights in older adults. For this reason, carrying out a study relating to this theme is warranted. Therefore, the aim of this study was to compare strength training protocols with elastic resistance and free weights on the muscle strength of knee extensors and flexors and on the functional performance of the lower limbs in the older adults. Our hypothesis is that elastic band training can alter muscle strength and functional performance in older adults, but it is not superior to free weight training.

2. Methods

2.1. Sample selection and randomization

This was a randomised clinical study carried out by applying strength training protocols with free loads and elastic resistance. The aim was to evaluate the effect on the muscle strength of the knee extensors and flexors and on the functional performance of the lower limbs in the older adults. All study protocols were approved by the Ethics and Research Committee, opinion No. 3.199.506, met the guidelines of the Declaration of Helsinki and registrem under No. RBR-5gqtbv in the Brazilian Registry of Clinical Trials (ReBEC). The interventions were carried out at gym Espaço Mais Saúde, located in the city of Monsenhor Paulo, Minas Gerais, Brazil. Prior to the test, the purpose and procedures of the study were explained to all eligible older adults, and they all gave their written consent to participate in the study.

The sample was selected for convenience, with 34 individuals (67 ± 5.25 years) with no previous experience in strength training. To meet the inclusion criteria, the older adults of both sexes, should be age 60 or over, able to practice physical activities and the proposed evaluations, be available to participate in the activities, absence of medical contraindications to the exercises, no self-reports of orthopedic injury to inferior limbs and without previous experience in strength training. The exclusion criteria were as follows: (1) refusal to participate in the study after inclusion, (2) symptoms or signs of orthopedic or metabolic alterations, (3) failure to attend the study protocol.

Three participants did not meet the inclusion criteria, then 31 individuals were selected and randomly assigned to two groups: TGFW (Training Group with Free Weights - 15 participants) and TGER (Training Group with Elastic Resistance - 16 participants). After the interview, randomization of volunteers was performed using a Random Number App version 2.1.0, by a trained researcher, who was responsible for the interventions.

2.2. Procedures

The tests were applied and performed in a single day for the execution of the first assessments of the participants. The first performance test was to sit and stand for 30 s. This was followed by the stationary gait for 2 min and culminated with the timed up and go test. All the tests were performed with a recovery interval of 1 min between them. A brief warm-up was performed prior to the MVIC test, and the participants performed knee extension and flexion on the respective machines. The peaks of force were measured by means of the load cell and by functional tests in terms of performance. On the following day, tests were conducted to measure elastic tension and the 10 RM strength test. All participants performed all these tests.

Three days after the evaluations, the application of the strength training protocol for the lower limbs was started, three times a week for eight weeks which were divided into four weeks of adaptation using 20% of the maximum load and the other four weeks with 75–80% of the maximum load (Mayer et al., 2011). Free weights were used for the TGFW group and elastic tubing was used for the TGER group. All the volunteers were able to complete 100% of the intervention, for a total of 24 resistance-training sessions.

At the end of the eight-week period, the extension test, knee flexion and functional performance were all repeated to compare the results. The assessment and intervention protocols are presented in Fig. 1.

2.3. Measuring instruments and measured variables

2.3.1. Sitting and standing test

A timer and a standard chair with a backrest, no arms and a seat height of approximately 50 cm were used. The participant was positioned sitting on the chair with his/her back resting on the back of the chair and their feet on the floor. The participant was encouraged to complete as many actions as possible of standing fully and sitting in 30 s. The score was obtained by the total number of correct executions in this time interval (Vicentini de Oliveira et al., 2020). The number of repetitions performed was recorded: The higher the value, the better the performance.

2.3.2. Stationary walking test

The use the 2-min stationary gait test, the responsible evaluator must choose and standardise for all volunteers: One knee is used as a reference (right or left side) to measure the number of elevations. The maximum number of knee elevations that the participant performed in 2 min was counted (Guedes et al., 2015).

2.3.2.1. Timed up and go test. The time needed to get up from the chair, walk 3 m as fast as possible, rotate 180°, move back and sit on the chair again (sit, walk, turn, and sit in the chair) was evaluated following the information available from Almajid et al. (2020). A stopwatch and a standard chair with a back, no arms, and a seat height of approximately 50 cm were used.

2.3.3. Assessment of maximum voluntary isometric contraction (MVIC)

Before the tests, the participants underwent a warm-up stage that consisted of 3 min on an ergometer bicycle (Embrex, Brusque, Brazil), with the maintenance of the heart rate (HR) at 50% of the HR value, maximum through the Xiaomi Mi Band 4 Smartwatch made in China (Monteiro et al., 2019).

For the verification of the isometric strength of the quadriceps, the participant assumed the sitting posture, with an erect trunk and hips

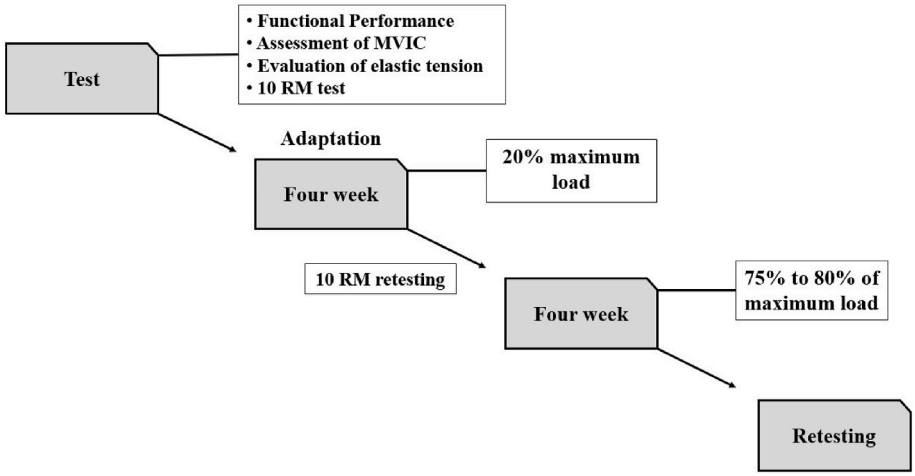

Fig. 1. Fluxogram of test and intervention.

flexed at 90° (Brown and Weir, 2001; Correa et al., 2011). The trunk was fixed by Velcro straps in a cross shape (McLeland et al., 2016). Then, the participant was placed in the prone position to assess the isometric strength of the hamstrings, with the lower limb to be assessed positioned in the distal region of the traction lever, with 70° of knee flexion. This angle was determined through knee goniometry (El-Ashker et al., 2017; Huang et al., 2017).

A portable digital dynamometer (E-sport, Brasília/DF), composed of a traction load cell with a maximum capacity of 200 kg of force, was used for the acquisition of the strength signals during the maximum voluntary isometric contraction (MVIC). The instrument was positioned next to the extending chair, perpendicular to the ground and connected to the cell phone of the Iphone 7 model, by Bluetooth 4.0 BLE, 2.4 G/hz communication, which transmitted the values of the muscle strength and performance acquired by graphics at the same time through the E-lastic 5.0 application (da Silva et al., 2021; Moura et al., 2020).

The procedure included the verbal and standardised incentive instructions provided before and during the test, with the command voice '1, 2, 3, already', the participant started the isometric contraction with the verbal stimulus 'strength, strength, strength, continues, strength, strength, strength, being encouraged to maintain maximum isometric strength and at the end of the limb-oriented test (Peek et al., 2018). This procedure was repeated for both lower limbs. The MVIC test was performed for 6 s with a 1-min interval between sets (three sets for each lower limb). At the end of the test, the average peak torque for each series applied was used.

2.3.4. Borg scale

To determine the training load to assess the intensity of the stimulus, the Borg scale of 0–10 points were applied at the end of each 10 RM test, where 0 means no effort and 10 means maximum effort (Vieira et al., 2019).

2.3.5. Evaluation of elastic tension

The evaluation of the elastic tension was measured using an analog scale with a hook, with a maximum capacity of 25 kg (Western brand FS-25 – 43874 Etilux, São Paulo, Brazil). For the evaluation of the elastic tension in each exercise, the elastics were fixed using Velcro straps at a fixed point on the backrest and at the other end of the same. An analogic scale was connected so the evaluator could measure the load imposed by the elastics. After reaching the required load for each participant, the evaluator marked the exact point on the floor where each volunteer should be positioned to perform the exercise. Specifically, sitting with the leg extended (final phase of the movement) coinciding with the point of tension measured by the scale. The participant's fixation point was determined on the malleolus of the leg of the person who would perform the movement; the position of the chair varied according to the tension of each of participant.

2.3.6. Test to determine the value of 10 RM

This test was performed according to the protocol of Vieira et al. (2020), where if the participant was not able to perform ten repetitions, the load was adjusted for the next attempt. Only three attempts were allowed, with a 3-min rest between them. Ten RM loads were obtained for all participants in two to three attempts, completing 10 repetitions in each series.

After a brief period of knee extensor and flexor stretching, in the knee extension movement (90 to 0°), the participant was positioned seated in a standardised chair 50 cm high, 44 cm wide by 40 cm. The hands were supported on the side of the chair and the participant performed the unilateral knee extension in the control of the evaluator (Brandão et al., 2020). The knee flexion (0–140°) was performed with the older adults participant standing close to the fixed back bar, which he/she held and used for support. In this test, the evaluator was near the participant and supported the respective knee to be evaluated with his forearm, thus providing security and support to the movement.

The tests were interrupted when the participants were unable to do the movement correctly (total range of motion without compensation in the execution of the movement). Trained professionals accompanied the older adults participants. Standardised verbal motivation was used during all periods of the tests.

2.4. Training protocol

During training, three sets were performed with 10 repetitions. There was a 60 s rest interval between the sets and the exercises (Fragala et al., 2019). The trainer used an attendance list for a frequency control of the participants.

The groups performed the knee extension and knee flexion exercises. In the training with elastic resistance, the distances between the patient and the place of fixation of the elastic resistance varied according to the calculation of the tension obtained with the 10 RM test. For the knee flexion exercise with elastic resistance, the participants performed the exercise standing up, supported with the same chair with which they performed the knee extension exercise in the sitting position, however, the chair position was reversed so that the individual would lean on and remain facing the backrest. The knee extension exercise was performed with the participant in a sitting position, with his/her back to the elastic fixation point.

2.5. Data analysis

The data were analysed for data normality using the Shapiro-Wilk test. Mean and standard deviation values were used to evaluate the descriptive data of the groups. The Mann-Whitney test was used for non-parametric data and the *t*-test for paired data. The independent *t*-sample test was used for the inter-group evaluation, and the *t*-test for dependent samples was used for the intra-group evaluation. The Mann-Whitney test was used in some cases of non-normality of the data for the inter-group evaluation, and the Wilcoxon test was used for the intra-group evaluation. For all tests, the significance value of $p < 0.05$ was considered and the statistical program GraphPad Prism version 8.0.1 was applied.

3. Results

After the eight weeks period, 29 individuals completed the intervention period with strength training. During the intervention process, two individuals were excluded of TGFW due to health issues. The fluxogram of the randomised clinical trial is shown in Fig. 2.

The characterisation of the sample, whose qualitative variables are described in Table 1, expressed as mean ± standard deviation and percentages, signals that there was a difference between the groups only when comparing age ($p \leq 0.05$). In the other variables, it can be said that there was homogeneity between the categories ($p > 0.05$). Participants were identified as physically active individuals according to the GPAQ: The minimum that they had to achieve in a typical week was 150 min of moderate physical activity, 75 min of vigorous physical activity, or a combination of the two possibilities.

Table 2 shows the results of intergroup and intragroup (pre and post strength training intervention) comparisons in physical performance. No significant differences were found between the groups and time.

Intergroup and intragroup comparison for the isometric muscle strength test analysed through the mean isometric peak torque in the knee extension and flexion movements of the dominant and non-dominant members. Significant differences were found only intergroup both pre- and post strength training intervention (Table 3).

Intergroup and intragroup comparison (pre- and post-strength training), and that only TGFW group achieved significant improvement in the 10 RM test in the extension and flexion movements. In the Borg scale of the flexion movement, there was a significant increase in the subjective perception of effort only in the TGFW group (Table 4).

At the end of the training protocol, the older adults participants in both groups reported a perception that the lower limbs were stronger (20.68%). There were reports of improvement in leg pain and the knee joints (44.82%), greater agility (31.03%) and better disposition when walking and climbing stairs (34.48%).

Table 1
Demographic data of study participants (n = 29).

Variables	TGFW (n = 13) Average ± DP	TGER (n = 16) Average ± DP	Value of p
Age (years)	70.0 ± 6.9	65.1 ± 3.6	0.006[a]
Height (cm)	154.2 ± 5.4	155.5 ± 5.5	0.54
Weight (kg)	70.5 ± 13.3	69.9 ± 13.1	0.90
BMI (kg/m)^2	29.6 ± 5.3	28.8 ± 4.4	0.64
GPAQ (min)	244.6 ± 44.2	273.1 ± 65.4	0.23
Sex (F/%)	84.6	93.7	
Sex (M/%)	15.4	6.3	

Abbreviations: TGFW - Training Group Free Weight; **TGER** - Training Group Elastic Resistance; **SD** - standard deviation; **n** - number of participants; **cm** - centimeters; **kg** - kilograms; **BMI** - Body Mass Index; **kg/m^2** - kilograms in square meters; **GPAQ** - Global Physical Activity Questionnaire; Min - minutes; **M** - Male; **F** - Female.

[a] p-value of the test for age with a significant difference $p < 0.05$, the non-parametric Mann Whitney test was used in the GPAQ evaluation and the unpaired *t*-test in the other evaluations.

4. Discussion

The analysis of isometric muscle strength demonstrated a significant difference in the peak isometric torque of the flexor muscles of the non-dominant leg intergroup and physical performance did not show any significant difference in the intra- and inter-group comparisons. Although the TGER group demonstrated a significant difference in the peak isometric torque of the flexor muscles of the non-dominant leg compared to the TGFW group. As this difference occurred both pre- and post-training, it demonstrates that the TGER group had a higher peak isometric torque of the flexor muscles of the non-dominant leg before starting training and this difference remained at the end of training.

A significant increase in training load was observed in the comparison between pre-training and after four weeks by the 10 RM test in the extension and flexion movements in the group TGFW. However, the intergroup evaluation demonstrated a significant difference in the pre-training period.

The finding of this study is partially reinforced by the clinical trial by Lima et al. (2018), where a conventional strength training (free weights) and a program using the elastic resistance with a total duration of 12

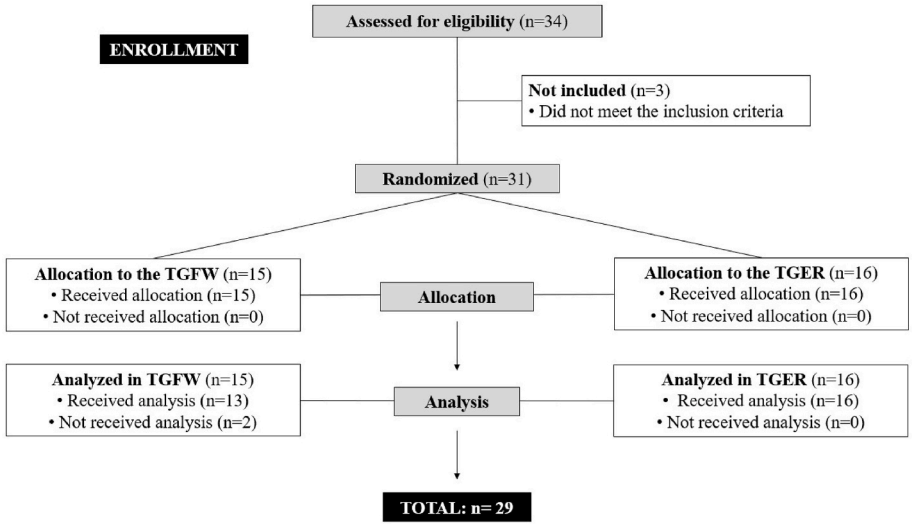

Fig. 2. CONSORT fluxogram of clinical study.

Table 2
Intergroup and Intragroup Comparison (pre- and post-training) of physical performance assessment in older adults after resistance training with free weights and elastic resistance.

	TGFW		TGER		Valor de p				
	Pre	Post	Pre	Post	G*T	T	G	Power	f^2
TSL 30 (rep)	12.3 ± 3.2	3.7 ± 3.21	13.3 ± 2.7	4.1 ± 2.1	0.53	0.04	0.47	0.17	0.37
TME2' (rep)	74.3 ± 16.0	72.7 ± 18.4	72.9 ± 17.0	75.0 ± 10.1	0.49	0.94	0.93	0.15	0.07
TUG (seg)	3.7 ± 0.7	3.7 ± 0.7	3.4 ± 0.4	3.4 ± 0.5	0.72	0.83	0.18	0.27	0.49

Abbreviations: TGFW - Training Group Free Weight; **TGER** - Training Group Elastic Resistance; **TSL 30** - Sit and Stand Test 30 s; **TME2'**- Stationary Walking Test 2 min; **rep** – repetition; **sec** - seconds **TUG** - Timed Up and Go Test. **G:** Groups; **T:** Time; **G*T:** Interaction Between Groups and Time. Values are presented as mean ± standard deviation.

Table 3
Intergroup and Intragroup comparison peak isometric torque values mean dominant and non-dominant leg.

	TGFW		TGER		Valor de p				
	Pre	Post	Pre	Post	G*T	T	G	Power	f^2
DLE (kg)	32.6 ± 8.0	37.1 ± 10.5	35.1 ± 11.2	33.8 ± 13.3	0.05	0.28	0.93	0.11	0.27
NDLE (kg)	32.4 ± 9.4	35.6 ± 11.3	34.1 ± 10.4	34.4 ± 10.9	0.40	0.32	0.95	0.06	0.10
DLF (kg)	9.6 ± 2.4	12.0 ± 3.6	12.0 ± 3.6	12.7 ± 4.7	0.94	0.44	0.03	0.07	0.16
NDLF (kg)	9.7 ± 3.1	10.1 ± 3.0	12.6 ± 3.0[&]	13.3 ± 4.1[&]	0.82	0.43	0.01	0.69	0.89

Abbreviations: TGFW - Training Group Free Weight; **TGER** - Training Group Elastic Resistance; **DLE**- Dominant Leg Extension; **NDLE** - Non-Dominant Leg Extension; **LDF** - Dominant Leg Flexion; **NDLF**- Non-Dominant Leg Flexion; **kg** - kilograms. **G:** Groups; **T:** Time; **G*T:** Interaction Between Groups and Time; Values are presented as mean ± standard deviation.
Note: **&** $p < 0.05$ difference intergroups in the same moment.

Table 4
– Intergroup and Intragroup comparison RM test values and Borg scale of the extensor and flexor.

	TGFW		TGER		Valor de p				
	Pre	Post	Pre	Post	G*T	T	G	Power	f^2
RME (kg)	6.8 ± 2.3	11.4 ± 5.0*	8.0 ± 1.6	13.0 ± 3.5	0.78	<0.01*	0.22	0.17	0.37
RMF (kg)	5.8 ± 1.9	8.0 ± 3.4*	6.5 ± 1.9[&]	9.3 ± 2.6	<0.01	0.48*	0.37	0.59	0.79
Borg Ext (index)	7.0 ± 1.6	7.5 ± 1.8	7.8 ± 1.1	8.2 ± 2.0	0.90	0.26	0.14	0.17	0.36
Borg Flex (index)	6.9 ± 1.4	8.6 ± 1.7*	7.3 ± 0.8	8.2 ± 1.5	0.23	<0.01	0.91	0.18	0.38

Abbreviations: TGFW - Training Group Free Weight; **TGER** - Training Group Elastic Resistance; **n** - number of participants; **SD** - standard deviation; **RME -** 10 RM test in the extension movement**; RMF -** 10 RM test in the flexion movement; **Borg Ext -** Borg Scale Test in the extension movement**; Borg Flex -** Borg Scale Test in the flexion movement**; kg -** kilograms. **G:** Groups; **T:** Time; **G*T:** Interaction Between Groups and Time; Values are presented as mean ± standard deviation.
Note: *$p < 0.05$ post vs pre (intragroup).
Note: **&:** $p < 0.05$ difference intergroups in the same moment.

weeks, were compared, and both training protocols were able to promote similar effects in healthy middle-aged adults to irregularly active older adults on muscle strength and functional capacity. As there was no difference between training with elastic resistance and training with conventional resistance (such as ankle weights and dumbbells). These findings allow professionals to opt for the use of elastic resistance, as such devices are available at low cost, are easy to use and can be used in different locations in day-to-day clinical interventions (Lopes et al., 2019b). However, the analysed studies included in this review also reported difficulties in controlling the intensity of exercises with elastic bands, with subjective scales of perceived exertion being applied. In this study, a manual scale was used, justifying its application by the ease in establishing intensity and its progression. Therefore, as it was not necessary to know the load imposed by the elastic given by the manufacturer, it was enough to move the distance between the fixed point and the moving point, which in our case was the chair, and combine the colors of the elastic to achieve the necessary load.

In the study conducted by Chen et al., in 2019, it was found that pre-frail elderly individuals who participated in an elastic exercise program for 8 weeks experienced an increase in grip strength and walking speed. The effects after 8 weeks were greater than those observed after 4 weeks. Additionally, the results indicated that women showed more significant improvements in grip strength compared to men. Therefore, elastic band exercises can be an effective strategy for enhancing muscle strength and walking capacity in pre-frail older adults, contributing to the prevention of progression to frailty states.

In support of the hypothesis, Martins et al. (2013) conducted an analysis on the effectiveness of progressive elastic resistance training programs for muscle strength in older adults classified as "healthy" and "unhealthy." The results indicated that elastic band training had significant effects on muscle strength in healthy older adults with some functional limitations, while it showed a moderate effect on muscle strength in older patients with pathologies. Furthermore, wheelchair-bound elderly residents of private nursing homes who received elastic resistance training achieved better functional fitness, daily life activities, and sleep quality than those who did not (Chen et al., 2016). In healthy adults, elastic resistance exercises have demonstrated effectiveness in improving functional performance and muscle strength. However, these exercises have not been shown to be superior to other resistance training methods for enhancing functional performance and muscle strength in healthy adults. As well as having a greater positive effect on women to stimulate changes in physical health and responses to steroid hormones (Oliveira et al., 2016; Furtado et al., 2020).

Load progression assessed by 10 RM tests increased significantly for the thigh extensor and flexor muscles, in TGFW group. Although it is done following the same criteria (same training protocol), the progression of the training load in both elastic and conventional groups may progress differently than others (Andersen et al., 2017). These effects

can be attributed to different factors that can make it difficult to control the progression of the training load. Since the deformation curve of the elastic components is associated with the increase in the values of force and tension, the total volume of the external load is not constant. Since there is no visual indicator of the total value generated in the tension, the load changes according to the extension of the elongation, these factors can make it difficult to assess the exact load for each exercise in daily clinical practice. Additionally, was observed by Silva et al. (2006), that a strength training program using 10 to 12 RM with a total duration of 12 weeks promotes increases in the muscular strength of the extensor and flexor muscles of the knees and elbows of older adults women, indicating that the prescription of strength exercises performed in zones of maximum repetitions is an interesting alternative to providing increases in muscle strength in older adults women.

Although the results do not present statistical differences for isometric muscle strength and physical performance in intergroup comparisons, an increase in isometric muscle strength values of knee flexors can be observed in the TGER group. There was also an increase in the number of repetitions with better results in the TGFW group. However, this increase did not change the performance of physical and functional tests.

At the end of the second assessment of isometric strength and functional performance, all study participants were asked about their individual perception of any improvement in quality of life and daily life after the training period. They reported the perception that the lower limbs were stronger, that there was an improvement in pain in the legs and knee joints, and greater agility and willingness when walking and climbing stairs.

Some strong points of this study need to be highlighted, which increase its relevance and make its results even more reliable. Firstly, this was a clinical study with a randomised homogeneous sample, which strengthens the degree of evidence found in the analyses. Secondly, it was possible to conduct a chronic intervention with planning for the adjustment of the training load, resulting in the need to adjust the prescribed intensity. It is known that the intensity of strength training is one of the most important variables and is directly linked to increases in strength and muscle mass in older adults (Fragala et al., 2019; Izquierdo et al., 2021). Thirdly, the possibility of investigating two variations of strength training enriches the literature on the topic and presents more options for clinical interventions to health professionals, since there were no differences between the modalities. Therefore, individuals who do not adapt to or do not have access to the practice of one of these modalities can benefit from the other, following correct prescription. Finally, this study standardised the training protocol using variables that make practice more accurate and safer for older adults such as volume, intensity, and intervals for rest between sets and exercises (Fragala et al., 2019; Izquierdo et al., 2021; Marzuca-Nassr et al., 2024; Schoenfeld et al., 2019).

Reported limitations of the study include the non-execution of a pilot study, to correct and previously identify possible flaws, and the lack of a control group, as it was a randomised clinical trial. Another limitation of this study was maintaining the intensity at 20% of the maximum load, which was intended to be used for a period of two weeks. However, because we were working with an untrained sample who had never participated in a strength training routine before, we chose to extend this intensity for an additional two weeks to ensure participant retention. We recognize the importance of applying progressive overload as one of the main strategies for strength gain, what was done in the final part of our research. We also acknowledge that maintaining this percentage of load may have influenced the lack of changes in the evaluated outcomes. Therefore, this decision was made to prevent dropout, as increasing the load could potentially lead to joint stress, pain, and reduced tolerance to effort (Fragala et al., 2019; Izquierdo et al., 2021). Furthermore, these same studies also advocate for the use of lower loads to achieve benefits from strength training. Another limiting factor was the intervention time, which was considered insufficient to assess the variables used. Eight weeks of elastic resistance training did not show significant changes in muscle mass and strength in untrained older adults (Martins et al., 2013).

5. Conclusion

TGFW was effective in the increasing the training load observed by the 10 RM test of the extension and flexion movements of the knee. As for the evaluation using the Borg scale, only the TGFW group presented greater difficulty in performing the knee flexion movement with the free weight. Both protocols not change the performance of physical and functional tests.

Clinical relevance

- Exercises with elastic resistance are a safe option for elderly individuals to engage, in addition to being cost-effective.
- Strength training with free weights and elastic resistance can facilitate access to exercise for older adults individuals.
- Strength training with elastic resistance and traditional strength training with free weights promoted the maintenance of muscular strength and functional performance in the elderly.

CRediT authorship contribution statement

Rafaela Zanin Ferreira: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Antonio Felipe Souza Gomes:** Writing – review & editing, Investigation, Conceptualization. **Marco Antonio Ferreira Baldim:** Writing – original draft, Methodology, Investigation. **Ricardo Silva Alves:** Writing - review & editing, Statistical analysis. **Leonardo César Carvalho:** Software, Formal analysis, Data curation. **Adriano Prado Simão:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Conceptualization.

Submission statement

The present manuscript has not been submitted, accepted, or published elsewhere.

Ethical approval statement

All study protocols were approved by the ethics committee of the Alfenas Federal University under opinion No. 3.199.506, met the guidelines of the Declaration of Helsinki and registered under No. RBR-5gqtbv in the Brazilian Registry of Clinical Trials (ReBEC).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbmt.2024.10.049.

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